

Large Language Models as Decision-Makers Under Uncertainty: Value of Information, Error Decomposition and an Extract-then-Solve Guardrail

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Abstract

[Intent: Problem (decisions under uncertainty handed to LLMs); the benchmark with exact Bayes-optimal actions on open-weight models; the two headline questions, value-of-information pricing (C5) and where the loss arises (C6); the guardrail (C4, C4b); the replications (C1, C2(b)); outcomes once recorded.]

Keywords: large language models, decision support, value of information, Bayesian decision theory, calibration, guardrails, benchmark

1. Introduction

[Intent: Motivate LLMs as decision aids in costly-test, triage, allocation and updating problems; state the gap (G1–G4 of the literature map); lead with C5 (do models price information?) and C6 (is the loss misreading, inference or action?); present C4 and C4b as the design answer; state C1 and C2(b) as replications of prior work on multi-action families and pinned open weights; list the contributions and the roadmap.]

2. Related work

[Intent: Position against LLM rationality and decision benchmarks (closest: Sun et al. 2026, exact regret with a belief/optimization split on binary tasks;

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Yamin et al. 2026, belief–action coherence), heuristics and biases in LLMs, calibration of stated probabilities, framing and order sensitivity, program-of-thought and tool use, and value of information in decision analysis; say what each covers and what Paper 8 adds.]

3. Benchmark: decision problems with exact optimal actions

[Intent: Define the four task families (F1 verification and stopping, F2 asymmetric-cost triage, F3 allocation under uncertain demand, F4 Bayesian updating), their exact solvers, normalized regret and the practical-optimality margin, trap and non-trap strata, decision-equivalent variants and the noise floor, the F1 balance on the test action, and the contamination controls, all fixed before any model run.]

3.1. Task families and exact solvers

3.2. Regret, strata and decision-equivalent variants

3.3. Item generation and contamination control

4. Models and decision conditions

[Intent: Describe the open-weight model panel (repositories, revisions, quantization, server, query dates) and the conditions: direct (P0), free reasoning (P1), structured direct (P2), program-of-thought (PoT), formula-given (FG) and the extract-then-solve guardrail (G), with decoding, parsing and sandbox rules. Describes the use of LLMs as the object of study (Elsevier: AI in research methods is described in Methods).]

4.1. Models

4.2. Conditions

4.3. Parsing, failures and checks by construction

5. Pre-specified analysis

[Intent: State the metrics and the reading rules (supports, weakens, refutes, inconclusive) fixed before any model run: VOI discrimination D (C5), the reading/inference/action regret decomposition and its order sensitivity (C6), guardrail comparisons (C4, C4b), and the replication metrics (C1, C1', C2, C3); bootstrap and interval methods; multiplicity; power at the run scale.]

- 5.1. *Value-of-information discrimination*
- 5.2. *Regret decomposition*
- 5.3. *Guardrail comparisons*
- 5.4. *Replications and secondary metrics*

6. Results

[Intent: Report each claim’s outcome as recorded, headline first: VOI pricing (C5), then the regret decomposition (C6), then the guardrail against direct, reasoning and program-of-thought (C4, C4b), then the replications (C1, C1’, C2, C3); per model and pooled.]

- 6.1. *Do models price information? (C5)*
- 6.2. *Where the loss arises: reading, inference or action (C6)*
- 6.3. *Extract-then-solve against direct, reasoning and program-of-thought (C4, C4b)*
- 6.4. *Replications: regret, traps, probabilities and invariance (C1, C1’, C2, C3)*

7. Discussion

[Intent: Implications for designing decision support around LLMs (what to hand to the model, what to hand to a solver); limitations (open-weight panel only, synthetic items, English prompts, decomposition order); threats to validity; how the framing changes under each recorded outcome, as stated in ARGUMENT.md.]

- 7.1. *Implications for decision support design*
- 7.2. *Limitations and threats to validity*

8. Conclusion

[Intent: Restate the question, the recorded outcomes of C5 and C6, the guardrail result, and what a practitioner should do.]

Data and code availability

[Intent: Generators, solvers, prompt templates, harness, item files (released after all runs), raw model responses, pinned model identifiers and result files, deposited on acceptance with a persistent identifier.]

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the writing process

[Intent: Owner’s statement on AI-assisted writing and coding, in Elsevier’s template wording.]

Appendix A. Prompt templates, schemas and reference solvers

[Intent: Give the fixed prompt templates per family and condition, the guardrail and P2 answer schemas, the formula-given sentences and the reference `decide()` functions.]

Appendix B. Checks by construction and generator diagnostics

[Intent: Report checks K1–K10 and the generator diagnostics (separation, strata and F1 test-action balance).]

References